A3-PS004-TC200

Authenticated Gateway-Assisted DNSSEC for IoT Clients

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Problem Description

- **Resource-constrained IoT devices** such as the ESP8266 face difficulties in performing complete DNSSEC validation due to limited memory and processing capabilities.
- **DNS responses** without proper validation may expose IoT devices to DNS spoofing and malicious redirection.
- **DNS queries transmitted** between the ESP8266 and the gateway require secure communication to prevent interception and tampering.
- **The device needs to authenticate** the DNS gateway to avoid trusting responses from unauthorized servers.
- **A reliable mechanism** is required to distinguish DNSSEC-valid, unsigned, and invalid DNS responses.
- **Gateway unavailability or authentication failure** must not result in an insecure plaintext DNS fallback.

The goal is to ensure secure DNS resolution for resource-constrained IoT devices.

Background

- **DNS Resolution:** DNS translates domain names into IP addresses, enabling IoT devices to communicate with online services.
- **DNSSEC Validation:** DNSSEC uses digital signatures to verify the authenticity and integrity of DNS data.
- **Resource Constraints:** Limited memory and processing capabilities make complete DNSSEC validation challenging for ESP8266 devices.
- **Gateway-Assisted Approach:** A trusted gateway performs DNS resolution and DNSSEC validation on behalf of the IoT device.
- **Secure Communication:** DNS-over-TLS protects DNS queries exchanged between the ESP8266 and the gateway.
- **Proposed Optimization:** The system aims to reduce client-side complexity while maintaining reliable DNS response validation.

Literature & RFC Review

- **DNS for IoT**: **A Survey (2023)** Reviews DNS requirements and security challenges in resource-constrained IoT environments.
- **Securing Name Resolution in the IoT: DNS over CoAP (2023)** Explores secure and efficient name resolution approaches for IoT devices.
- **CSProp: Ciphertext and Signature Propagation Low-Overhead Public-Key Cryptosystem for IoT Environments (2021)** Investigates reducing public-key cryptographic overhead in constrained environments.
- **Analysis of Robust and Secure DNS Protocols for IoT Devices (2025):** Examines the security and robustness of DNS protocols used by IoT devices.
- **RFC 4033 & RFC 4035:** DNSSEC security requirements and protocol validation.
- **RFC 7858:** DNS over TLS (DoT) for encrypted DNS communication.
- **Research Gap:** Resource-constrained IoT clients need secure DNS resolution without the overhead of performing full DNSSEC validation locally.
- **Our Approach:** Gateway-assisted DNSSEC validation using Unbound, encrypted client–gateway communication, and ESP8266-side response classification.

Proposed Solution

- **ESP8266 Client:** The NodeMCU ESP8266 generates DNS queries for domain names and sends them to the configured gateway.
- **Secure DNS Communication:** DNS queries are transmitted over DNS-over-TLS (DoT) to protect communication between the ESP8266 and the gateway.
- **Gateway-Based Validation:** Unbound acts as the recursive resolver and performs DNSSEC validation before returning DNS responses.
- **Authoritative Test Zones:** NSD hosts controlled SECURE, INSECURE and BOGUS test zones to evaluate different DNSSEC validation outcomes.
- **Response Classification:** The ESP8266 analyses DNS response codes and the AD flag to distinguish valid, unsigned and invalid DNS responses.
- **Gateway Authentication:** Public-key pinning and HMAC-based channel binding are intended to verify gateway identity and prevent unauthorized access.
- **Failure Handling:** The design aims to reject failed secure connections and avoid falling back to plaintext DNS.

Optimization Proposed

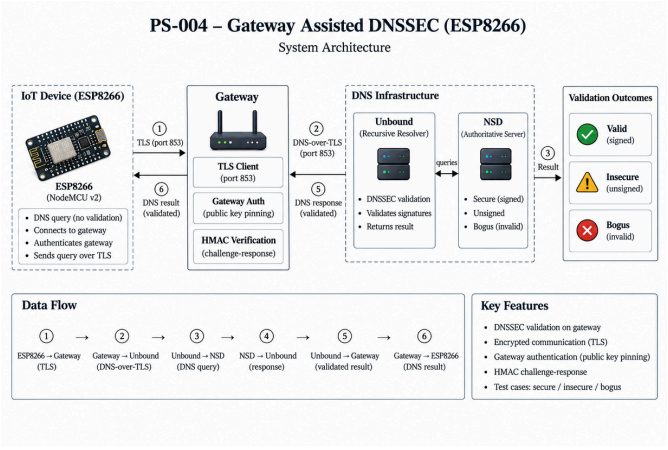
Aspect	Conventional Approach	Proposed Approach
DNSSEC Validation	Validation may be difficult on resource-constrained devices.	DNSSEC validation is delegated to the gateway.
Device Workload	Client handles more DNS processing tasks.	Gateway handles DNS resolution and validation.
DNS Communication	Queries may lack transport encryption.	DNS-over-TLS protects client-to-gateway traffic.
Gateway Trust	Server identity verification depends on configuration.	Public-key pinning and HMAC-based channel binding are planned.

Optimization Proposed

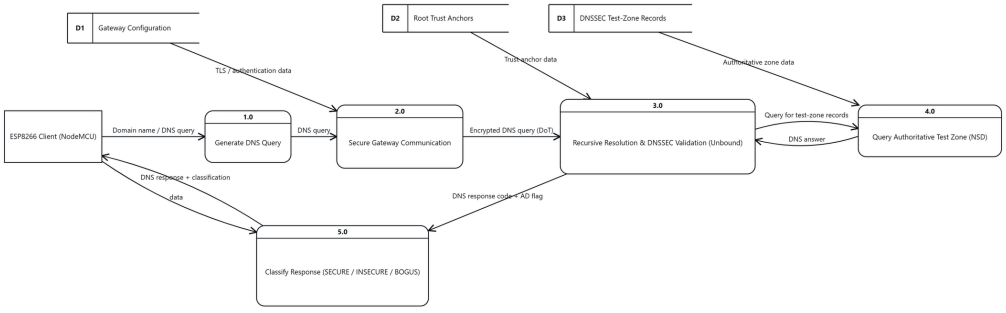
Aspect	Conventional Approach	Proposed Approach
Response Verification	DNS responses may be accepted without clear classification.	Responses are classified as SECURE, INSECURE or BOGUS.
Failure Handling	Fallback behavior depends on implementation.	The design aims to reject failed secure connections without plaintext fallback.

Optimization Goal: Reduce ESP8266-side complexity while improving DNS communication security and response validation.

Solution Architecture & Design



Data Flow Diagram



Deployment Timeline

Milestone 1 — Environment Setup Configured the ESP8266 development environment using PlatformIO and established the Ubuntu WSL environment for NSD and Unbound deployment.
Milestone 2 — DNSSEC Infrastructure Created SECURE, INSECURE and BOGUS test zones using NSD and configured Unbound for recursive DNS resolution and DNSSEC validation. Verified the expected DNS response behaviour.
Milestone 3 — Client Development Implemented ESP8266 Wi-Fi connectivity, DNS query construction and DNS response parsing to interpret response codes and the AD flag.
Milestone 4 — Gateway Integration Configured gateway-side DNS-over-TLS connectivity and tested network communication between the ESP8266 and the Unbound resolver. TLS handshake issues remain under investigation.
Milestone 5 — Security Validation Planned verification of gateway public-key pinning, HMAC-based channel binding, response classification and secure failure handling.

Work Methodology

Develop → Integrate → Validate → Demonstrate

Develop the client and DNSSEC infrastructure, integrate the ESP8266 with the gateway, validate DNS responses and security controls, and demonstrate the complete system using the three DNSSEC test zones.

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Thank You !